

INTERNSHIP PRESENTATION

Python Developer Internship

AMAN RAJ

Roll No. 25SCS1003003193 • Section 2CSE15

04 WEEKS

31 JULY — 31 AUGUST 2026

CODEC

Technologies • Python Internship

Four weeks strengthened practical Python skills

The internship progressed from Python fundamentals and advanced concepts to data handling, APIs and industry-oriented project work.

100%

Portal completion

04

Learning weeks

02

Project briefs

Host: Codec Technologies • Role: Python Intern • July–August 2026

STUDENT PROFILE

**Aman
Raj**

Roll number

25SCS1003003193

Section

2CSE15

Institution

IILM University, Greater Noida

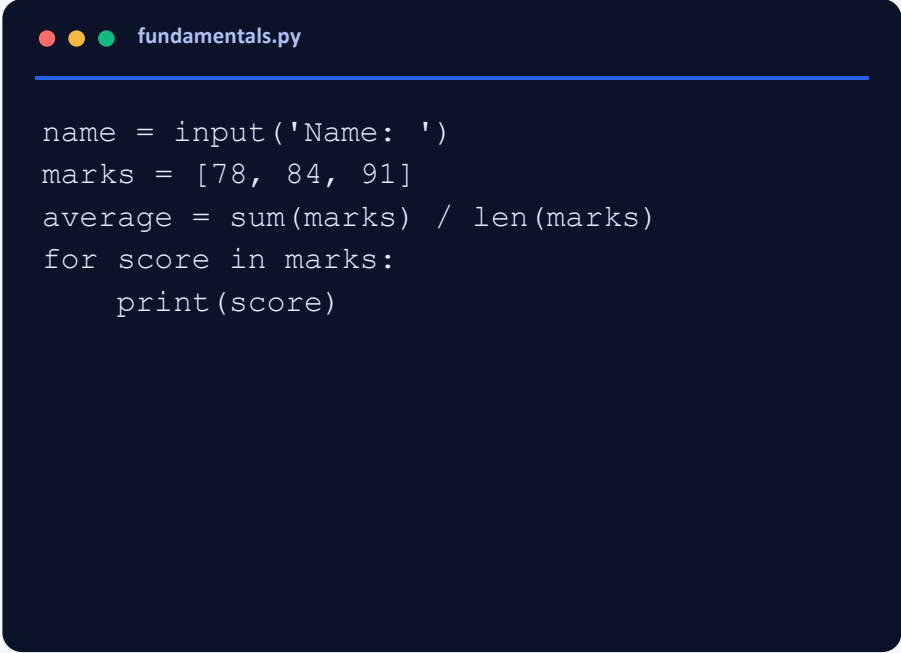
Five milestones built an end-to-end Python workflow

The curriculum moved from core syntax and reusable functions to OOP, data processing, APIs and project implementation.

**LEARNING LOGIC**

Fundamentals → modular code → data → APIs → project delivery

Python fundamentals established a reliable coding base

fundamentals.py

```
name = input('Name: ')
marks = [78, 84, 91]
average = sum(marks) / len(marks)
for score in marks:
    print(score)
```

Core practices

- Use variables and built-in data types correctly
- Apply conditions and loops to control program flow
- Write reusable functions with clear inputs and outputs
- Organize code for readability and simple debugging

EXPECTED RESULT

Small command-line programs that solve structured problems with predictable results.

Advanced Python made programs easier to maintain

Object-oriented design, file operations and exception handling improved structure and reliability.

- OOP** — Reusable classes and objects
- FILES** — Read and write persistent data
- ERRORS** — Safer execution with try and except
- MODULES** — Organized code through imports

student_record.py

```
class Student:
    def __init__(self, name):
        self.name = name
try:
    record = Student('Aman')
except ValueError as error: ...
```

OUTCOME • Programs became easier to extend, reuse and troubleshoot

Data handling turned records into insight

```
data_analysis.py

import pandas as pd
df = pd.read_csv('records.csv')
df = df.dropna()
df['score'] = df['score'].astype(int)
summary = df.groupby('course').mean()
```

Data workflow

- Load structured data from CSV and JSON
- Clean missing or inconsistent values
- Transform rows with reusable functions
- Summarize results for reporting or export

KEY LEARNING • Validate data before drawing conclusions or saving output

API practice connected Python to live services

The learning module introduced web requests, JSON responses and lightweight endpoints for application integration.

01

SELECT

Endpoint and parameters chosen

02

REQUEST

HTTP request sent

03

CHECK

Status and errors validated

04

PROCESS

JSON data transformed

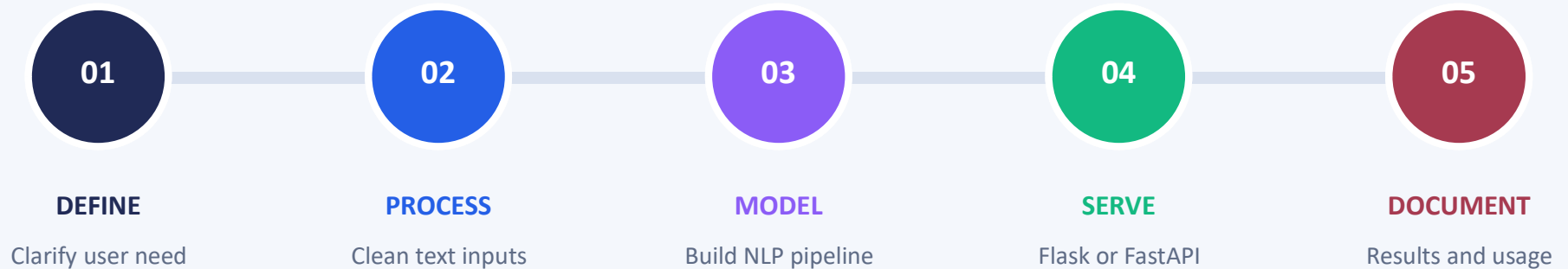
api_client.py

```
response = requests.get(url, timeout=10)
response.raise_for_status()
payload = response.json()
return payload
```

OUTPUT • Structured API data made available to a Python application

Two project briefs combined Python skills

The final stage applied Python, NLP, document processing and lightweight web frameworks to practical application concepts.



PROJECT FOCUS • AI-powered chatbot | Automated resume parser

The toolkit covered the Python workflow

FOUNDATION

Core Python

Syntax • data types • control flow • functions

DESIGN

Advanced

OOP • files • modules • exception handling

DATA

Analysis

Pandas • CSV • JSON • SQL and SQLite

DELIVERY

Web and NLP

Requests • Flask/FastAPI • NLTK/spaCy • GitHub

Tools were practiced as connected layers, from clean code to usable applications.

A repeatable workflow linked problem to delivery

This workflow reflects the progression from understanding a requirement to validating and sharing a Python solution.



PRINCIPLE • Build, test, then integrate the solution

Practice targeted common Python project risks

The modules highlighted practical issues that arise when scripts move from examples to reusable applications.

CHALLENGE	TECHNICAL RESPONSE	INTENDED EFFECT
INPUTS VARY	Validate types, formats and missing values	Fewer avoidable processing failures
LOGIC GROWS	Split work into functions, classes and modules	Easier maintenance and testing
APIS FAIL	Add timeouts, status checks and exceptions	More reliable service integration
OUTPUTS DRIFT	Use test cases and clear result formats	Consistent, explainable behavior

These responses summarize the curriculum and project requirements; no unsupported performance metrics are claimed.

Four weeks show growing Python capability

The portal records 100% course completion and concludes with two industry-oriented project requirements.

01

Foundation scripts

Syntax, conditions, loops, functions

02

Modular programs

OOP, files, exceptions, modules

03

Data workflows

Pandas, CSV, JSON, SQL

04

API integration

Requests, JSON, Flask/FastAPI

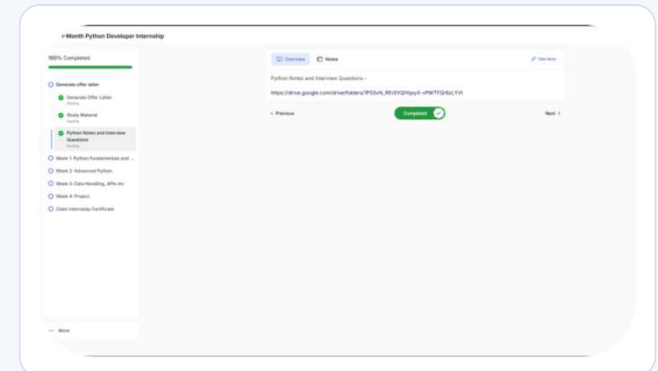
05

Project delivery

Chatbot and resume-parser concepts

100% COMPLETE

Attached records confirm internship completion



100%

INTERNSHIP PROGRESS

The certificate records Aman Raj's one-month Python Developer Internship at Codec Technologies.

The internship built technical delivery skills

TECHNICAL DEPTH

Build Python solutions with clarity and confidence

- Write readable, modular Python
- Work with files and structured data
- Consume APIs and validate responses
- Apply NLP and document-processing concepts
- Present and document project outcomes

PROFESSIONAL HABITS

01 Progressive problem solving

Break requirements into clear, testable steps.

02 Self-paced learning

Complete structured modules independently.

03 Quality awareness

Use validation, exceptions and clear outputs.

04 Professional communication

Document work and prepare it for professional sharing.

CONCLUSION

Ready to apply a complete Python development workflow

The internship connected Python fundamentals, modular programming, data handling, APIs and project planning into one practical learning journey.

AMAN RAJ • 25SCS1003003193 • 2CSE15

NEXT APPLICATION

01

Build larger Python projects

02

Strengthen testing skills

03

Publish documented work

Acknowledgement

With gratitude to IILM University and Codec Technologies for enabling this practical learning experience.